SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

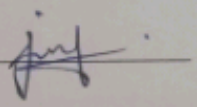
COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I Jannathul Firdaus.N (192573025) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Scenario-Based Questions

SET 2:

1.A retail company collects terabytes of customer transaction data every day from online and offline stores. The management wants to analyze purchasing patterns in near real time to improve product recommendations and inventory management.

- Design a cloud-based big data architecture using distributed storage and processing frameworks.
- Explain how batch processing and stream processing can be integrated.
- Discuss how data lakes and data warehouses can be used together.
- Recommend techniques to ensure scalability and cost efficiency.

2.A healthcare organization is migrating patient records and medical imaging data to the cloud. The solution must comply with strict privacy regulations while allowing authorized doctors to access records securely from multiple locations.

- Design a secure cloud architecture that protects sensitive healthcare data.

- Explain how encryption, identity and access management (IAM), and key management should be implemented.
- Discuss auditing, logging, and compliance monitoring strategies.
- Recommend methods to prevent unauthorized access and data breaches.

3.An online ticket booking platform experiences sudden spikes in traffic whenever popular events are announced. The company wants to reduce infrastructure management while ensuring fast response times.

- Design a serverless cloud architecture for the application.
- Explain how event-driven computing can improve scalability.
- Discuss the role of managed databases, API gateways, and message queues.
- Evaluate the advantages and limitations of serverless computing for this workload.

4.A multinational company gathers data from sales, finance, manufacturing, and customer support systems. The organization requires a centralized analytics platform with high-quality and consistent data.

- Design a cloud-based ETL/ELT pipeline for integrating data from multiple sources.
- Explain how data quality, metadata management, and governance should be maintained.
- Discuss the role of workflow orchestration and scheduling tools.
- Recommend strategies for optimizing query performance and storage costs.

5.A logistics company wants to predict delivery delays using historical shipment data, weather information, and live GPS tracking. The analytics platform must support continuous model training and real-time predictions.

- Design a cloud-based machine learning and analytics architecture.
- Explain how big data technologies can support model training and inference.
- Discuss the use of cloud-native AI/ML services for model deployment and monitoring.
- Recommend strategies for handling large-scale data ingestion, feature engineering, and model updates.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

ig Data Architecture for Retail Analytics.

Cloud - Based Big Data Architecture

A retail company can use cloud - based Big Data technologies to analyze large volumes of customer transaction data collected from online and offline stores.

Data Ingestion: Apache Kafka collects transaction data from different sources.

Distributed Storage: Cloud storage services such as Amazon S3 store raw and processed data.

Data processing: Apache Spark processes large dataset using distributed computing.

Analytics: Cloud data warehouse and visualize tool help identify purchasing patterns and improve product recommendation.

Data Lakes and Data Warehouses

A data Lake stores large amounts of raw data in different formats including transaction records and customer activity data.

A data warehouse stores cleaned and structured data for reporting and analysis.

The company can store raw data in the data lake processes it and transfer the

required information to data warehouse.

Integration of Batch and stream processing

Batch processing: process historical sales data at scheduled intervals to identify long-term purchasing trends.

Stream processing: Analyzes live transactions continuously to identify current customer preferences

Integration: Apache Spark supports batch processing while ~~Spark~~ Apache Kafka and Apache Flink can support real time data processing.

Scalability and cost Efficiency:

Auto-scaling: Automatically adjusts computing resources according to demand.

Data compression: Reduces storage requirements.

Storage Tiering: Moves older data to lower-cost storage.

Serverless processing: Reduces costs by charging according to resources usage

Resource monitoring: Identifies unnecessary resource consumption.

...ure cloud Architecture for Healthcare Data.

Secure cloud Architecture

A healthcare organization must protect patient records and medical images while allowing authorized doctors to access them securely from different locations.

The architecture should include:

cloud Storage: Stores patient records and medical images securely.

Private networks: Protect communication between hospitals and cloud services.

Identity and Access Management (IAM): Ensures that only authorized users can access patient information.

Backup and recovery: maintains backups to recover data after accidental deletion or system failure.

Access control: gives doctors and healthcare staff permissions according to their roles.

Encryption, IAM and Key Management:

Encryption:

- Use encryption at rest to protect stored patient records.
- Use TLS encryption to secure data transmitted between hospitals and cloud services.
- Encryption prevents unauthorized users from reading sensitive information.

Identity and Access Management (IAM):

- Implement role-based access control
- Use multi-factor authentication to verify user identities.
- Follow the principle of least privilege granting users only the permissions necessary for their duties.
- Review access permissions regularly.

Key management:

- Use a cloud key management services to securely manage encryption keys.
- Restrict access to key to authorized users and services.
- Rotate key periodically and monitor their usage.

Auditing, Logging and compliance Monitoring.

Audit logs: Record who accessed patient records and what actions they performed.

Activity Monitoring: Track login attempts file downloads and permission changes.

Automated alerts: Detect suspicious activities and repeated failed login attempts.

Compliance monitoring: Regularly check security configurations against applicable healthcare privacy regulations.

Log Protection: Prevent unauthorized modification or deletion of audit records.

Preventing Unauthorized Access and Data Breaches:

* Use multi factor authentication for user accounts
* Apply role based access control and least privilege permission.
* Protect networks using firewalls and network segmentation.
* Regularly update software and apply security patches.
* Maintain encrypted backups and test recovery procedures.

3. Serverless Architecture for an Online Ticket platform:

Serverless cloud Architecture:

A serverless architecture allows the ticket booking company to run its application without directly managing servers.

The architecture include:

API gateway: Receives booking requests from web and mobile applications

Serverless Functions: AWS Lambda or Azure function process ticket searches, bookings and cancellation.

Managed Database: Stores event details, ticket availability and customer information

Message Queue: Handle background tasks such as sending booking confirmations.

Monitoring tools: Track application performance and errors.

Event driven computing and Scalability:

* Functions automatically scale according to incoming requests subjects to service limits
* Message queues manage sudden increases in requests.
* Background tasks can run asynchronously without delaying the main booking process.

Role of Managed Databases, API gateways and Message Queues:

Managed Databases: Store booking information and ticket availability while reducing database maintenance work.

API gateways: Provide secure access to application services and support authentication and request throttling.

Message Queues: Manage large volume of tasks and separate background processing from critical booking operations.

Advantages and Limitations:

Advantages	Limitations
Automatic Scaling during traffic spikes.	Cold starts can increase response time.
Reduced server maintenance	Function execution limits may affect long running tasks
Pay per use billing	costs may increase unexpectedly during heavy traffic
Faster application development	Debugging distributed applications can be difficult.
Easy integration with event driven services.	Dependence on a specific cloud provider may cause vendor lock in.

4. Cloud - Based ETL / ELT pipeline for Ente.

Data Integration:

Design a cloud - Based ETL/ELT pipeline.

A multinational company can integrate data from sales, finance, manufacturing and customer support systems using a centralized cloud analytics platform.

Extraction: collect data from databases applications, APIs and files.

Ingestion: Transfer the collected data to cloud storage.

Transformation: clean, validate, standardize and combine data from different sources

Analytics: Use business intelligence tools to generate reports and dashboards.

Data Quality, Metadata Management and governance

Data Quality:

* Remove duplicate records
* Identify missing and incorrect values
* Standardize dates, currencies and formats
* validate data before processing.

Metadata Management:

* Maintains information about data source formats and ownership.
* Use a data catalog to help users find and understand datasets.
* Track data lineage to understand how data is transformed.

Data governance:

* Define data ownership and access permissions
* Apply data security and privacy policies
* Establish retention policies and conduct regular audits.

Role of workflow orchestration and scheduling tools:

* Schedule data processing jobs
* Execute tasks in the correct order
* Manage dependencies between tasks.
* Retry failed jobs automatically.
* monitor pipeline execution and generate alerts

Query performance and storage cost Optimization

Partitioning: Divides large datasets by date or region to improve query performance

Compression: Reduce Storage requirements.

Columnar Formats: formats such as parquet improve analytical processing efficiency.

Query optimization: Avoids unnecessary columns and processes only required data.

Storage Tiering: Moves infrequently accessed data to lower-cost storage.

Resource Scaling: Adjusts computing resources according to workload.

5. Cloud-Based Machine Learning Architecture for Predicting Delivery Delays.

A logistics company can use cloud-based machine learning to predict delivery delays using historical shipment data weather information and live GPS tracking.

The architecture includes:

* Data ingestion: Collect shipment records, weather updates and live GPS data.
* Cloud Storage: Store raw and processed data in a cloud data lake.
* Data processing: Use Apache Spark to process large datasets.
* Feature engineering: Generate features such as distance remaining,

weather conditions and historical delivery times.

* Model training: Train machine learning models using historical data.
* Model Deployment: Deploy the trained model as a cloud-based prediction service.
* Real time prediction: Analyze live data to estimate delivery times and identify possible delays.
* monitoring: Track model accuracy and prediction performance.

Role of Big Data Technologies in Training and Inference:

Model Training:

* Apache Spark processes large volume of data.
* Distributed computing speeds up data preparation and model training.
* Historical GPS, weather, shipment records help identify patterns associated with delivery delays.

Model Inference:

* Stream processing tools analyze live GPS and weather updates.
* The deployed model uses updated data to predict possible delays.
* Predicts are shared with logistics managers to support timely decisions.

Cloud Native AI/ML Services for Model deployment and monitoring:

* Model Training provides managed computing resources for training.
* Model Deployment hosts trained models for real time predictions.
* Auto scaling adjusts computing resources according to demand.
* monitoring track prediction accuracy and changes in incoming data.
* Version control maintains model versions and allow rollback when required.

Data Ingestion, Feature Engineering and Updates.

* Use Apache Kafka to collect live bus data Obtain weather information using API. Store historical and incoming data in cloud storage.
* calculate distance remaining and estimated travel time. combine weather condition with shipment location analyze delivery patterns.
* Monitors accuracy to identify performance degration. automation data preparation, model training and evaluation. Test updated models before deployment.